

# APEX METROLOGY & CALIBRATION LABS

## Equipment Performance Verification & Inspection Checksheet

|                  |                                             |                    |                             |
|------------------|---------------------------------------------|--------------------|-----------------------------|
| Instrument Name: | High-Precision Analytical Spectrophotometer | Model / Serial No: | X-Series 4000 / SN-884211-B |
| Location / Dept: | Quality Control Laboratory - Bay 3          | Inspection Date:   |                             |
| Inspected By:    |                                             | Next Due Date:     |                             |

**General Instructions:** Ensure the instrument has completed its initial warm-up cycle before recording values. Measure each parameter against the standard reference values and verify whether the recorded value falls within the permitted operational range. Circle or check **PASS** or **FAIL** accordingly.

| REFERENCE | PARAMETER                          | VALUE      | RANGE             | PASS / FAIL                                                 |
|-----------|------------------------------------|------------|-------------------|-------------------------------------------------------------|
| 01        | Baseline Voltage Stability         | _____ V    | 4.95V - 5.05V     | <input type="checkbox"/> Pass <input type="checkbox"/> Fail |
| 02        | Wavelength Accuracy (Deuterium D2) | _____ nm   | 656.1nm ± 0.2nm   | <input type="checkbox"/> Pass <input type="checkbox"/> Fail |
| 03        | Photometric Linearity (at 1.0 ABS) | _____ ABS  | 0.995 - 1.005 ABS | <input type="checkbox"/> Pass <input type="checkbox"/> Fail |
| 04        | Stray Light Threshold (at 220 nm)  | _____ %T   | ≤ 0.020 %T        | <input type="checkbox"/> Pass <input type="checkbox"/> Fail |
| 05        | Optical Alignment / Lamp Energy    | _____ mA   | 45.0mA - 52.0mA   | <input type="checkbox"/> Pass <input type="checkbox"/> Fail |
| 06        | Chamber Core Temperature           | _____ °C   | 23.5°C ± 1.5°C    | <input type="checkbox"/> Pass <input type="checkbox"/> Fail |
| 07        | Cooling Fan Noise / Speed          | _____ RPM  | 2400 - 2800 RPM   | <input type="checkbox"/> Pass <input type="checkbox"/> Fail |
| 08        | Signal-to-Noise Ratio (RMS)        | _____ dB   | ≥ 60.0 dB         | <input type="checkbox"/> Pass <input type="checkbox"/> Fail |
| 09        | Zero-Point Drift (over 30 mins)    | _____ ABS  | ≤ 0.001 ABS/hr    | <input type="checkbox"/> Pass <input type="checkbox"/> Fail |
| 10        | Interface/RS232 Data Transmit Rate | _____ kbps | 9.6kbps ± 0.1kbps | <input type="checkbox"/> Pass <input type="checkbox"/> Fail |

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Technician Signature / Date

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Quality Assurance Reviewer / Date